Tutorial 5 - A Brief Revision on C Language

Q5.1 Which of the following lines of C Code is implemented to detect the button press of an **Active Low** input connected to STM32F103 pin PB_0 and instantiated as button_PB0 (i.e., DigitalIn button_PB0 (PB_0))?

(a) while (button_PB0 == 0);
(b) while (button_PB0 = 0);
(c) while (button_PB0 = 1);
(d) while (button_PB0 == 1);



Q5.2 Which of the following lines of C-code shifts the value of the variable “count” by 2 bits?

(a) count = count + 2;
(b) count = count * 2;
(c) count = count < 2;
(d) count = count >> 2;



Q5.3 What is the value of the variable “output_u8” after the code below is executed?

```
…
unsigned char output_u8 = 232;

output_u8 = output_u8 & 0x55;
…
```

(a) 0b01000000
(b) 0b10000100
(c) 0b00100010
(d) 0b11001000



Q5.4 Based on the C code below,

unsigned char value_u8 = 0x54;

(a) what is the number base format of the constant value used to initialize the variable “value_u8”? hexadecimal
(b) rewrite the code above to initialize the variable “value_u8” using C code representation for the following number bases.

(i) Decimal
(ii) Binary

(i) = 0x84
(ii) = 0xb01010100

Q5.5 Based on the C code below, what is the final value of the variable “output_u8” below after the code has been executed completely?

```
unsigned char output_u8 = 0x40;
output_u8 = output_u8 | 0x01;
```

0x40 = 0b01000000

0x40 | 0x01 = 0b11111111

Q5.6 Characters can be compared using their ASCII code values. For instance, the ASCII code for ‘A’ is 65, for ‘B’ is 66, and for ‘a’ is 97. So, ‘B’ > ‘A’ because 66 > 65 and ‘a’ > ‘A’ because 97 > 65. Write your C code to find the “Greatest” letter in the word “Singapore”.

```
#include <stdio.h>
#include <string.h>

int main (){
        char word[] = "Singapore";
        int length = strlen(word);
        char greatest = word[0];

for (int i = 1; i < len ; i++){
        if (word[i] > greatest){
        greatest = word[i];
    }
 }

printf("Greatest letter: %c\n", greatest)
printf("ASCI value: %d\n", greatest)

return 0;
}
```